SOFTWARE REQUIREMENTS SPECIFICATION

DOCUMENTATION

JB-Link Telco©
Stay Connected, Stay Ahead.

Table of Contents

1. Introduction
2. Purpose
3. Functionality
4. Non-Functionality
5. User Interface
6. Constraints
7. Appendices
8. Glossary

1. Introduction

This document details the complete software requirements specification (SRS) for JB Link Telco. It describes the purpose, scope, definitions, system overview, functional and non-functional requirements, system features, and external interface requirements of the system. This document is intended for use by the development team, stakeholders, and any other parties involved in the software development lifecycle.

Company Branding Placeholder:

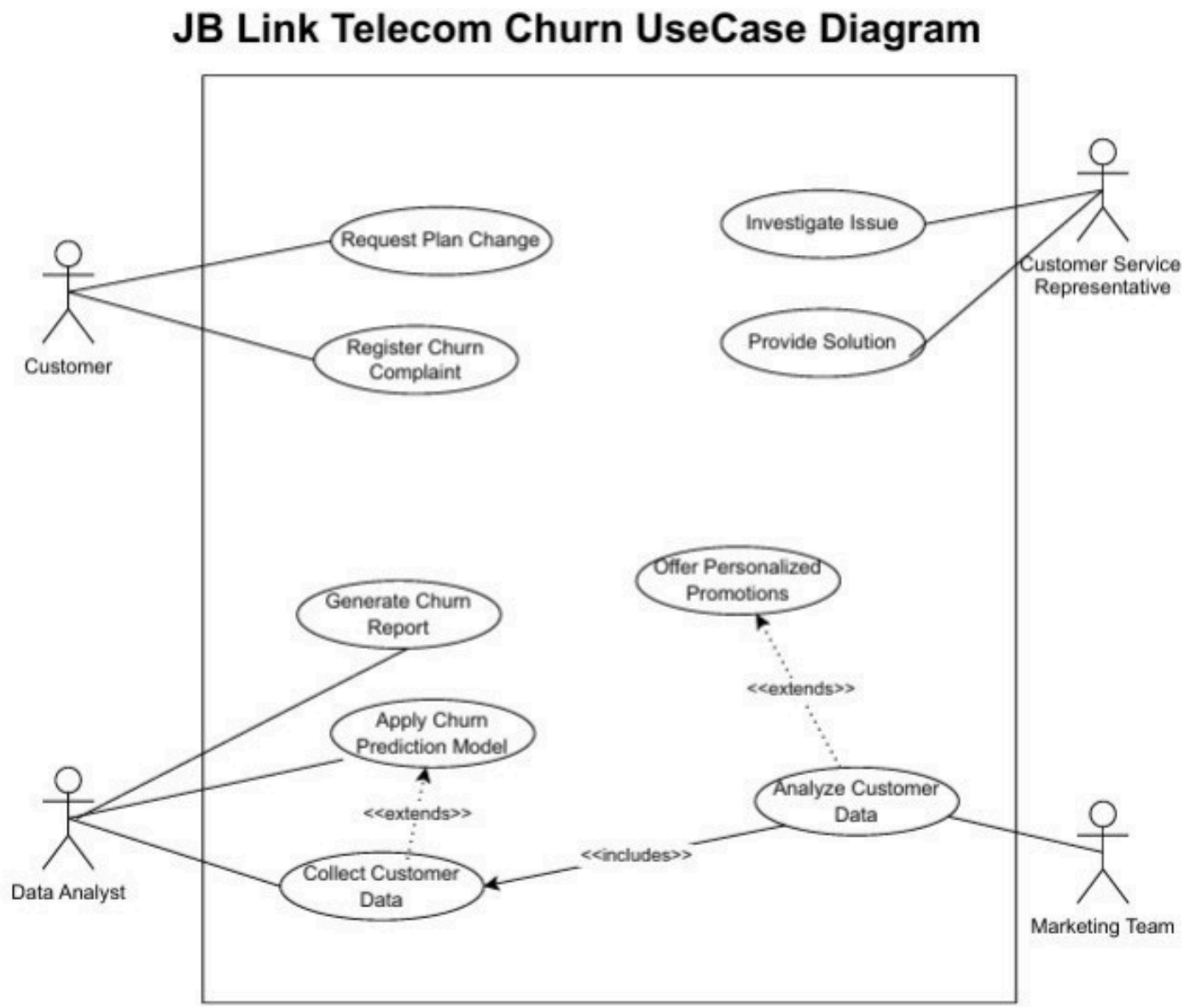
JB-Link Telco©
Stay Connected, Stay Ahead.

2. Purpose

The purpose of this software is to develop a predictive churn analytics system for a telecommunications company to identify customers likely to churn. This system will aid in taking preventive actions using machine learning models applied to historical customer data.

3. Functionality

- Accepts customer data (demographics, services, usage, etc.) for processing.
- Applies a trained churn prediction model to determine churn risk.
- Visualizes results through a dashboard, highlighting high-risk customers.
- Allows users to filter, sort, and download reports.
- Sends alerts or flags customers above a churn risk threshold.
- Tracks model accuracy and retraining timelines.



4. Non-Functionality

Performance: System must respond to churn queries within 2 seconds.

Scalability: Should handle a dataset of at least 10,000 customers.

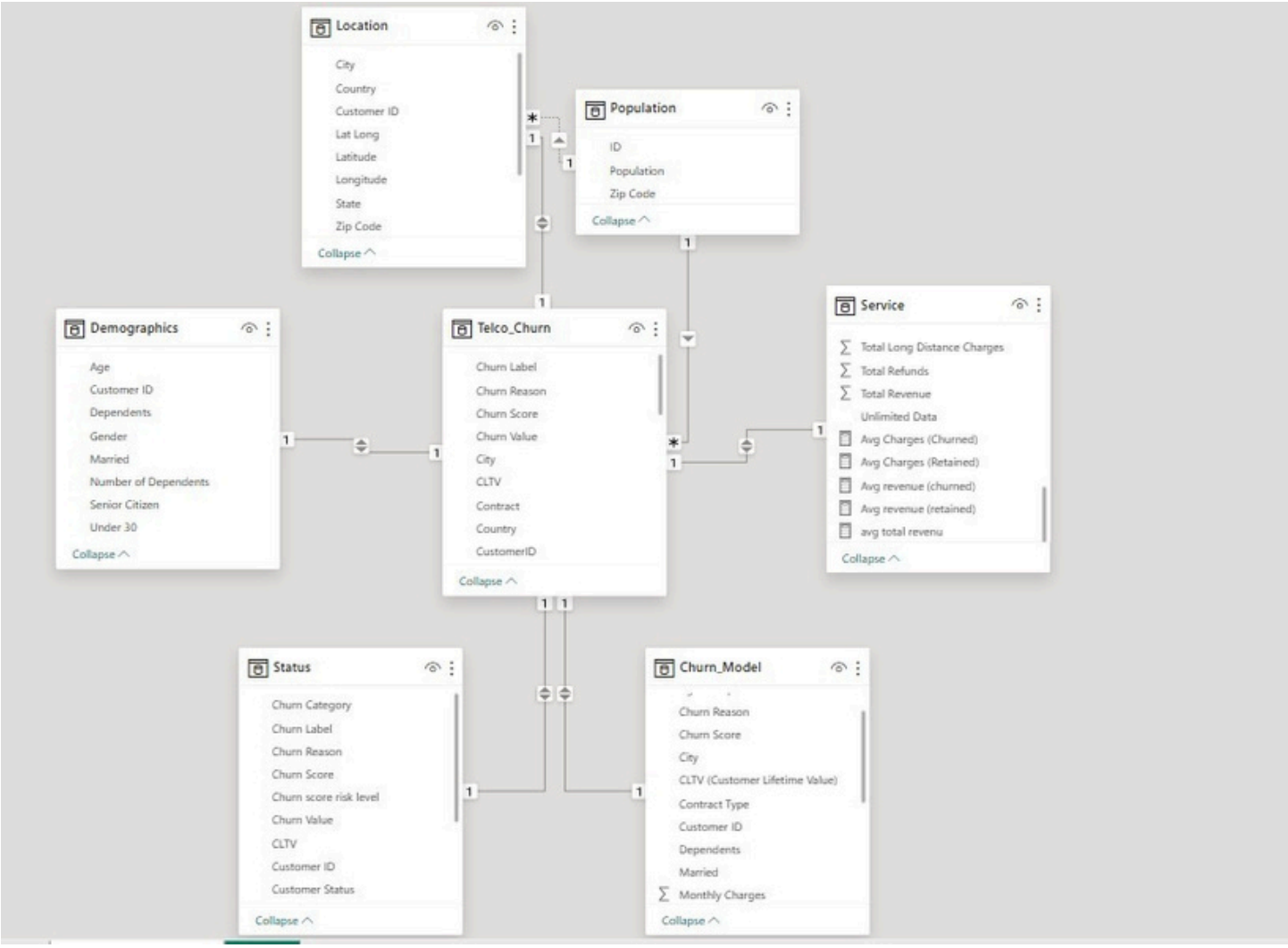
Availability: 99.5% uptime expected during business hours.

Security: Role-based access to data and dashboard.

Maintainability: Code must be modular and allow for future model updates.

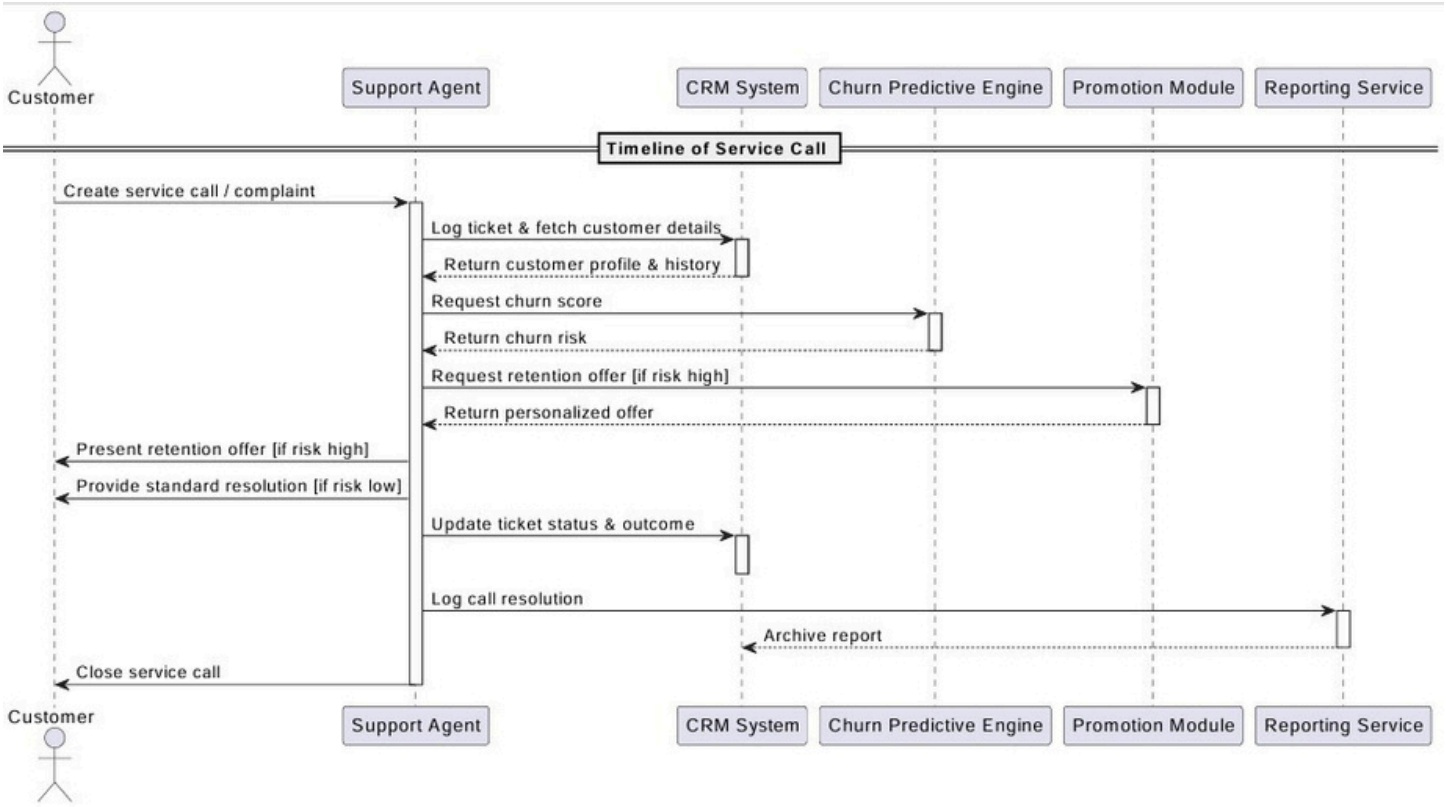
5. Data Requirements

- **Source:** Data from internal CRM systems and billing platforms; the Kaggle dataset used for training.
- **Type:** Numeric and categorical (e.g., tenure, payment method, contract type).
- **Quality Expectations:**
  - No more than 5% missing values.
  - Consistent format across all categorical fields.
  - Validated values (e.g., monthly charges must be positive).
- **Frequency:** Batch uploads (daily); real-time processing for new entries optional.
- **Storage:** SQL database for structured storage and retrieval.



6. User Interface

- **Framework:** Web-based UI built using React.js or equivalent.
- **Components:**
  - Summary Dashboard: KPIs like churn rate, total customers, high-risk customers.
  - Customer Table View: Search, filter, and drill-down functionality.
  - Model Accuracy Panel: View F1 score, AUC, and last retrain date.
  - Export: PDF/CSV report download option.



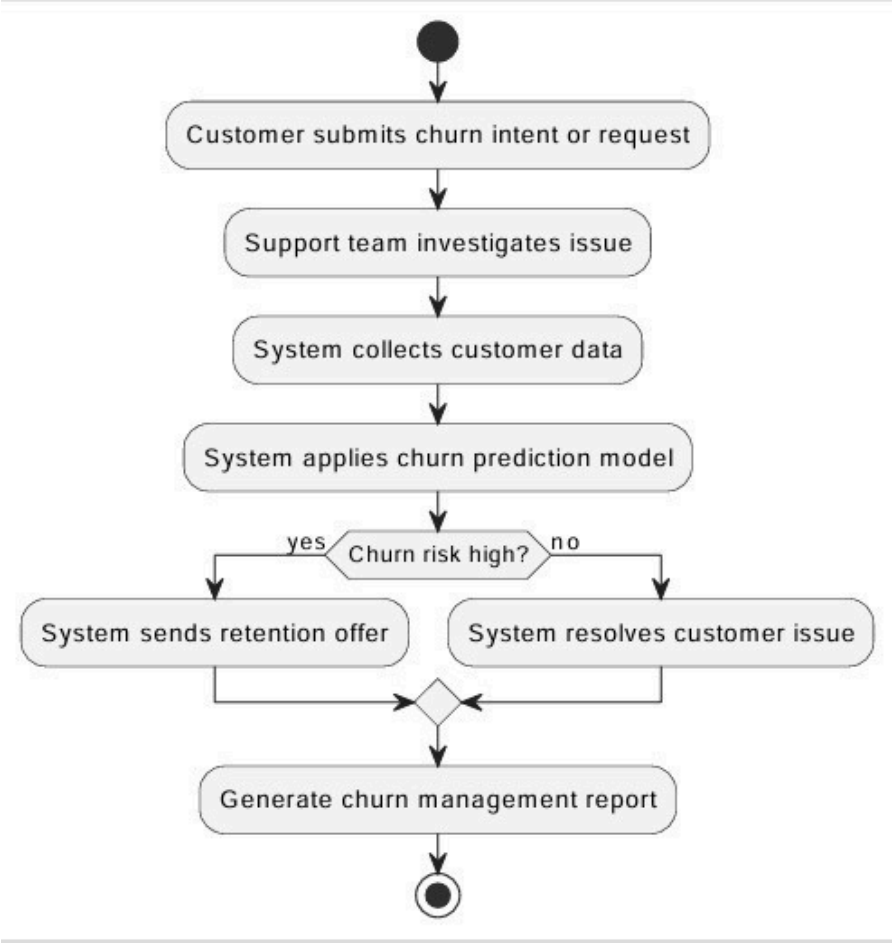
7. Constraints

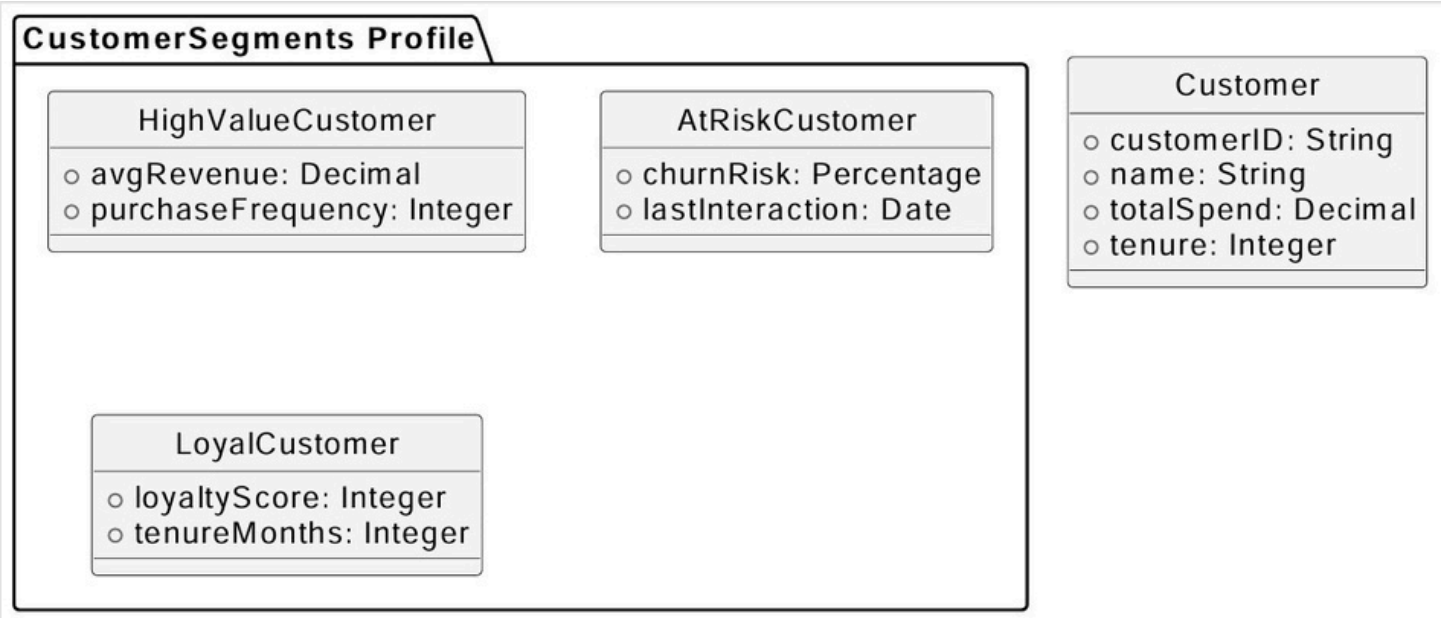
- The backend database **must** be implemented using **SQL** (MySQL/PostgreSQL).
- The system must comply with data privacy norms (e.g., GDPR/India PDP Bill).
- Only internal datasets are to be used for the first phase (no API integrations).

8. Appendices

Sample Report Structure:

- Customer ID | Churn Probability | Risk Level | Contract Type | Monthly Charges
- Export Format: CSV and PDF
- Visual Charts: Pie chart for churn distribution, bar chart for churn by service type.


Customer submits churn intent or request
Support team investigates issue
System collects customer data
System applies churn prediction model
Churn risk high?
yes
no
System sends retention offer
System resolves customer issue
Generate churn management report


CustomerSegments Profile
HighValueCustomer
avgRevenue: Decimal
purchaseFrequency: Integer
AtRiskCustomer
churnRisk: Percentage
lastInteraction: Date
Customer
customerID: String
name: String
totalSpend: Decimal
tenure: Integer
LoyalCustomer
loyaltyScore: Integer
tenureMonths: Integer

9. Glossary

Term	Explanation
Churn	When a customer discontinues services with the company.
MVP	Minimum Viable Product – first usable version of the system.
AUC	Area Under Curve – a model evaluation metric.
CRM	Customer Relationship Management – system storing customer data.
SQL	Structured Query Language – used to manage relational databases.